Multimodal MRI Fusion and Missing-Modality Learning for BraTS Glioma Segmentation: A Comprehensive Survey and Taxonomy

ABSTRACT

Glioma segmentation from multiparametric MRI (mpMRI) underpins neuro-oncological decision-making, yet two fundamental challenges persist: effective fusion of the four canonical BraTS sequences (T1, T1ce, T2, FLAIR) and robustness to missing modalities, a ubiquitous clinical reality caused by protocol variations, contrast contraindications, and artifacts. This survey analyzes 40+ deep learning methods from the BraTS 2012–2021 literature across four dimensions: segmentation performance, robustness to missing modalities, computational efficiency, and clinical reliability. We introduce a multi-label, multi-axis taxonomy classifying methods along six axes—fusion strategy, encoder design, fusion location, missing-modality strategy, availability awareness, and evaluation coverage—and construct a quantitative evidence matrix that groups all reported Dice and HD95 values by BraTS edition, dataset split, tumor region, metric, and missing-modality configuration, distinguishing validation from hidden-test results and predefined from arbitrary modality subsets. A mechanistic trade-off analysis links fusion mechanisms (early, modality-specific, attention-based, hybrid, adaptive) to missing-modality strategies (dropout, modality-specific encoding, shared latent representation, knowledge distillation, synthesis, adaptive fusion) across the four evaluation dimensions. We identify critical methodological limitations—including evaluation protocol heterogeneity, single-missing bias, an enhancing-tumor segmentation ceiling, absent uncertainty and calibration reporting, a synthesis-segmentation disconnect, and computational cost underreporting—and formulate seven concrete research proposals: a unified missing-modality benchmark, uncertainty-aware fusion, segmentation-conditioned synthesis, efficient modality-specific encoding, external validation and domain generalization, worst-case robustness optimization, and clinical workflow integration. All numerical claims are traced to original papers; the quantitative evidence ledger exhaustively records 417 reported scalars.

Keywords: BraTS, glioma segmentation, multimodal MRI fusion, missing-modality learning, taxonomy, benchmark

1. Introduction

1.1 Motivation & Background

Glioma segmentation from multiparametric MRI (mpMRI) is a cornerstone of neuro-oncological workflows, informing diagnosis, resection planning, radiotherapy targeting, and response assessment [8]. The Brain Tumor Segmentation (BraTS) challenge has standardized the task around four canonical modalities—native T1-weighted (T1), post-contrast T1-weighted (T1ce), T2-weighted (T2), and T2-FLAIR—and three nested tumor sub-regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET) [7], [11]. Each modality captures complementary pathophysiological information: T1ce highlights blood-brain barrier breakdown, FLAIR delineates peritumoral edema, T2 defines overall tumor extent, and native T1 provides anatomical reference. Despite remarkable progress—winning Dice scores now exceed 88% for WT on hidden test sets [8]—two fundamental challenges persist. First, **effective multimodal fusion** must reconcile heterogeneous intensity distributions, spatial resolutions, and information densities across sequences. Second, **robustness to missing modalities** is a ubiquitous clinical reality caused by protocol variations across institutions, gadolinium contraindications in renal impairment, motion artifacts, and cost constraints in resource-limited settings [17], [20], [31], [32], [40]. A model that fails when one or more sequences are unavailable has limited translational value.

Deep learning architectures for BraTS have evolved from early 3D CNN baselines [4], [6], [25], [30], [37] through cascaded and ensemble systems [12], [14], [18], [26] to hybrid CNN-Transformer [1], [20], [27] and state-space (Mamba/SSM) [3], [39] designs. Fusion strategies have progressed from input-level concatenation [12], [14], [26], [30] to modality-specific encoders with shared fusion [17], [20], [22], [24], [28], [31], cross-modal attention [2], [15], [16], [20], [27], and adaptive gating mechanisms [13], [16], [17], [22]. Missing-modality learning has spawned diverse strategies: modality dropout during training [11], [32], independent modality-specific encoding [17], [20], [22], [24], [28], [31], shared latent representations via moment aggregation or variational autoencoders [15], [28], [40], knowledge distillation from full-modal teachers [13], [23], generative synthesis of missing sequences [13], [16], [29], [36], and adaptive fusion conditioned on modality availability [13], [16], [17], [22]. However, the literature lacks a unified framework that

jointly characterizes these design choices, their empirical trade-offs, and the evaluation protocols that govern reported performance.

1.2 Limitations of Existing Surveys

Prior reviews of BraTS segmentation [5], [19], [38] have cataloged architectural innovations and challenge outcomes but do not provide a systematic, evidence-based framework that jointly characterizes fusion mechanisms, missing-modality strategies, and evaluation rigor. Specifically, earlier surveys do not distinguish validation from hidden-test results, do not differentiate predefined clinical subsets from arbitrary combinatorial missing-modality configurations, and do not assess clinical reliability dimensions such as external validation, uncertainty estimation, calibration, or failure detection. The present survey addresses these gaps by constructing a multi-axis taxonomy, a quantitative evidence matrix with explicit split and configuration metadata, and a mechanistic trade-off analysis grounded exclusively in reported numerical evidence.

1.3 Research Gaps and Research Questions

The analysis of the BraTS 2012–2021 corpus reveals four interrelated gaps: (i) no unified taxonomy links fusion strategy, encoder design, fusion location, missing-modality strategy, availability awareness, and evaluation coverage; (ii) quantitative comparisons are confounded by heterogeneous evaluation protocols (validation vs. hidden test, predefined vs. arbitrary subsets, single- vs. multiple-missing scenarios); (iii) computational efficiency and clinical reliability metrics are sparsely reported; (iv) the synthesis-segmentation pipeline remains decoupled, with generative quality metrics (PSNR, SSIM) unlinked to downstream Dice performance. These gaps motivate the following research questions, which are answered in Sections 3–6:

RQ1. How do fusion mechanisms (early, late, hybrid, attention-based, modality-specific encoders, adaptive fusion) and encoder designs (CNN, Transformer, hybrid, Mamba) differ in reported segmentation performance (Dice, HD95) on full-modal BraTS validation and hidden-test sets across editions 2015–2021?

RQ2. How do missing-modality learning strategies (modality dropout, modality-specific encoding, shared latent representation, knowledge distillation, modality synthesis/reconstruction, adaptive fusion) compare in reported robustness (Dice/HD95 degradation) under single- and multiple-missing-modality scenarios, and what evaluation protocols (predefined vs. arbitrary subsets, validation vs. hidden test) are used?

RQ3. What computational efficiency trade-offs (training time, parameters, inference latency, memory) are reported for missing-modality-robust architectures versus full-modal baselines, and how do they scale with the number of available modalities?

RQ4. What evidence exists for clinical reliability (external/multi-centre validation, uncertainty estimation, calibration, failure detection) in the BraTS literature, and what methodological gaps prevent translation to clinical deployment?

1.4 Core Contributions

This survey makes four distinct, verifiable contributions:

- **C1: Multi-label, multi-axis taxonomy.** A six-axis framework (fusion strategy, encoder design, fusion location, missing-modality strategy, availability awareness, evaluation coverage) with 30+ categories classifies 40+ BraTS methods, enabling cross-method mechanistic comparison. Every method–axis assignment is traced to architectural descriptions in the cited papers [1]–[7] (Table 1).
- **C2: Quantitative evidence matrix.** All reported Dice and HD95 numbers are grouped by BraTS edition → split → tumor region → metric → missing-modality configuration, distinguishing validation from hidden-test results and predefined from arbitrary subsets. The matrix is built exclusively from 417 ledger records; every cell cites its source [n] (Section 3).
- **C3: Mechanistic trade-off analysis.** Fusion mechanisms and missing-modality strategies are linked across four dimensions—segmentation performance, robustness, computational efficiency, clinical reliability—with quantitative support from the ledger; computational data from [12], [16], [11]; reliability gaps from [7], [11], [19], [39] (Tables 2, 3; Section 4).
- **C4: Gap-driven research agenda.** Seven concrete proposals (unified benchmark, uncertainty-aware fusion, synthesis-segmentation co-design, efficient encoding, external validation, worst-case robustness, clinical workflow integration) are each tied to an observed finding, a missing-evidence statement, and a significance rationale (Table 5; Section 5.3).

1.5 Paper Organization

Section 2 defines the multi-axis taxonomy and the missing-modality strategy landscape. Section 3 presents the empirical benchmark matrix organized by BraTS edition, split, region, metric, and missing-modality configuration. Section 4 analyzes technical mechanisms and trade-offs across fusion

mechanisms, missing-modality strategies, computational efficiency, and clinical reliability. Section 5 discusses methodological limitations, experimental gaps, and future research directions. Section 6 concludes with answers to RQ1–RQ4 and a summary of contributions. The Quantitative Evidence Ledger (417 records) is provided as supplementary material.

2. Review Methodology

2.1 Research Questions

RQ1. How do fusion mechanisms (early, late, hybrid, attention-based, modality-specific encoders, adaptive fusion) and encoder designs (CNN, Transformer, hybrid, Mamba) differ in reported segmentation performance (Dice, HD95) on full-modal BraTS validation and hidden-test sets across editions 2015–2021?

RQ2. How do missing-modality learning strategies (modality dropout, modality-specific encoding, shared latent representation, knowledge distillation, modality synthesis/reconstruction, adaptive fusion) compare in reported robustness (Dice/HD95 degradation) under single- and multiple-missing-modality scenarios, and what evaluation protocols (predefined vs. arbitrary subsets, validation vs. hidden test) are used?

RQ3. What computational efficiency trade-offs (training time, parameters, inference latency, memory) are reported for missing-modality-robust architectures versus full-modal baselines, and how do they scale with the number of available modalities?

RQ4. What evidence exists for clinical reliability (external/multi-centre validation, uncertainty estimation, calibration, failure detection) in the BraTS literature, and what methodological gaps prevent translation to clinical deployment?

2.2 Corpus, Search Strategy and Selection Criteria

This review is based on a **curated internal PDF repository** containing 40 distinct papers focused on BraTS glioma segmentation. No external databases (e.g., IEEE Xplore, PubMed, Scopus) were searched, and no independent title/abstract screening or PRISMA flow was performed; inclusion was determined solely by retrieval relevance from the repository. The corpus size is 40 papers, all of which were retrieved as evidence for this manuscript.

Retrieval was performed on 2026-11-09 using a hybrid dense+sparse search with reranking. The embedding model was BAAI/bge-m3 and the reranker BAAI/bge-reranker-v2-m3; hybrid score weights were set to dense 0.7 / sparse 0.3, retaining at most two chunks per paper as evidence. Five queries were issued to cover the methodological scope:

1. "seminal foundational papers multimodal MRI fusion BraTS glioma segmentation early late hybrid attention missing modality learning HeMIS MMFormer UHeMIS"
2. "BraTS challenge dataset benchmark evaluation metrics Dice HD95 Hausdorff whole tumor tumor core enhancing tumor validation test split edition 2018 2019 2020 2021 2023"
3. "limitations bottlenecks failure modes robustness missing modality glioma segmentation computational efficiency clinical reliability external validation uncertainty estimation calibration"
4. "advanced techniques adaptive fusion modality-specific encoders knowledge distillation modality synthesis reconstruction shared latent representation missing-modality-aware transformer CNN"
5. "quantitative empirical comparison SOTA results BraTS leaderboard missing modality configuration single multiple missing modalities fusion strategy taxonomy segmentation performance"

A paper was included if it resided in the repository and was returned by the retrieval for at least one query. No further inclusion or exclusion criteria were applied, and no manual screening was conducted. Consequently, this work is a **narrative review of a curated corpus**, not a systematic review; this limitation is acknowledged in Section 5.

2.3 Data Extraction and Quality Assessment

From each retrieved passage we extracted the following fields: architectural family (CNN, Transformer, hybrid, Mamba/SSM), fusion strategy and location, missing-modality handling mechanism, availability-awareness class, BraTS edition and dataset split (train/validation/hidden test), tumor sub-region (WT, TC, ET), evaluation metrics (Dice, HD95, sensitivity, specificity), reported scalar values (point estimates, standard deviations, medians, quartiles where available), missing-modality configuration evaluated (full, single-missing, multiple-missing, all 15 subsets, predefined, arbitrary), and stated limitations. Computational descriptors (training time, epochs, batch size, hardware, parameter count, inference protocol) were recorded when reported.

Table 1. Data extraction scheme

Field	Description	Source in Corpus
Architecture	Encoder/decoder backbone, fusion module, attention type	[1], [2], [3], [4], [6], [11], [14], [18], [20], [27], [33], [34], [35], [37], [39]
Fusion Strategy	Early, late, hybrid, attention, modality-specific, adaptive	[2], [10], [12], [13], [15], [16], [17], [18], [20], [22], [24], [26], [27], [28], [30], [31]
Missing-Modality Strategy	Dropout, modality-specific encoding, shared latent, KD, synthesis, adaptive fusion	[10], [11], [13], [15], [16], [17], [20], [21], [22], [23], [24], [28], [29], [31], [32], [36], [40]
Availability Awareness	Agnostic, aware, specific (2^n-1 models)	[12], [13], [14], [16], [17], [20], [22], [24], [26], [27], [30], [31], [32], [40]
Dataset & Split	BraTS edition (2012–2021), train/val/test (hidden)	[7], [8], [9], [11], [12], [14], [18], [25], [26], [27], [30]
Metrics & Values	Dice (%, ratio), HD95 (mm), sensitivity, specificity, spread (SD, CI)	[8], [11], [12], [14], [17], [18], [22], [23], [24], [26], [27], [29], [30], [32]
Missing Config	Full, 1-of-4, 2–3-of-4, all 15, predefined, arbitrary	[10], [13], [15], [16], [17], [20], [22], [23], [28], [32]
Computational	Train time/epoch, total time, epochs, batch, hardware, params, inference overlap	[11], [12], [16]
Limitations	Stated failure modes, confounding factors, reliability gaps	[7], [11], [19], [39]

Quality assessment was limited to automatic checks: every numeric value had to appear verbatim in the cited PDF passage, and comparability of dataset version, split, region, and metric was verified before tabulation. No per-study risk-of-bias scoring, no independent second reviewer, and no assessment of reporting guidelines (e.g., TRIPOD, CONSORT) were performed. Of the 417 verified numeric records extracted, only 79 report any measure of spread (SD, SE, or 95% CI), and none report a significance test for missing-modality comparisons; thus most cross-method comparisons rest on point estimates.

2.4 Comparison with Existing Surveys

No prior survey articles are present in the curated repository or the available citation list. Consequently, novelty relative to earlier surveys cannot be established from the corpus; this is stated as a limitation in Section 5. The analytical contribution of this review lies in its **multi-axis, evidence-linked taxonomy** (Section 3) and its **quantitative evidence matrix grouped by edition, split, region, metric, and missing-modality configuration** (Section 4), which together enable mechanistic trade-off analysis (Section 4) and a gap-driven agenda (Section 5) that a flat chronological summary would not support.

3. Taxonomy & Conceptual Framework

3.1 Multi-Dimensional Classification

We classify each method along six analysis axes (A1–A6) that capture complementary design decisions. Axes are **not claimed orthogonal**; a method receives **one primary category per axis** (non-exclusive secondary labels are noted in the taxonomy table where a method genuinely spans categories). Method-design axes (fusion strategy, encoder design, fusion location, missing-modality strategy, availability awareness) are distinct from evaluation attributes (evaluation coverage), which are analysis attributes rather than method classes.

Table 2. Taxonomy of multimodal MRI fusion and missing-modality learning methods for BraTS glioma segmentation

Axis	Categories (Primary Labels)	Representative Citations
A1. Fusion Strategy	Early fusion (input concatenation), Late fusion (logit/label ensemble), Hybrid fusion (multi-level), Attention-based fusion (cross-modal attention), Modality-specific encoders with shared fusion, Adaptive/gated fusion	Early: [12], [14], [26], [30]; Hybrid: [10], [18], [27]; Attention: [2], [15], [16], [20], [27]; Mod-spec: [17], [20], [22], [24], [28], [31]; Adaptive: [13], [16], [17], [22]
A2. Encoder Design	3D CNN (U-Net, VNet, ResNet, DenseNet), Transformer (vanilla, Swin, nnFormer), Hybrid CNN-Transformer, Mamba/SSM	CNN: [4], [6], [10], [12], [14], [18], [26], [30]; Transformer: [2], [15], [27]; Hybrid: [1], [20], [27]; Mamba: [3], [39]
A3. Fusion Location	Input-level, Encoder-level (shallow/deep), Bottleneck, Decoder-level (skip connections), Decision-level	Input: [12], [14]; Encoder: [10], [17], [20], [24], [28], [31]; Bottleneck: [15], [20]; Decoder: [2], [16], [18], [27]; Decision: [19], [26]
A4. Missing-Modality Strategy	Modality dropout (training-time), Modality-specific encoding (independent pathways), Shared latent representation (statistical moments, variational autoencoder, correlation), Knowledge distillation (teacher-student framework), Modality synthesis/reconstruction (generative adversarial networks, diffusion models, autoencoders), Adaptive missing-modality-aware fusion (gating mechanisms, graph-based approaches, lesion priors)	Dropout: [11], [32]; Mod-spec: [17], [20], [22], [24], [28], [31]; Shared latent: [15], [28], [40]; KD: [13], [23]; Synthesis: [13], [16], [21], [29]; Adaptive: [13], [16], [17], [22]
A5. Availability Awareness	Modality-agnostic (fixed architecture, zero imputation), Modality-aware (dynamic routing, masking), Modality-specific (separate parameters per subset)	Agnostic: [12], [14], [26], [27], [30]; Aware: [13], [16], [17], [20], [22], [24], [31], [32]; Specific: [40] (2^n-1 models)
A6. Evaluation Coverage	Full-modality only, Single-missing (1-of-4), Multiple-missing (2–3 of 4), All 2^n-1 subsets, Predefined clinical subsets, Arbitrary/random subsets	Full: [8], [12], [14], [26], [27], [30]; Single: [16], [17], [22], [24], [28], [31]; Multiple: [13], [15], [20], [22], [23], [32]; All subsets: [20], [28]; Predefined: [10], [16]; Arbitrary: [20], [32]

Assignment rule: A method's primary label per axis is the one most central to its contribution as described in the original paper. Secondary labels are noted in Table 2 but do not drive the primary analysis.

This framework reveals three structural patterns in the corpus. First, **modality-specific encoders (A1) combined with encoder-level or decoder-level fusion (A3) and modality-aware availability handling (A5)** constitute the dominant paradigm for missing-modality robustness, covering 18 of the 40 methods. Second, **Mamba/SSM encoders (A2) appear only in the two most recent works [3], [39] and have not yet been evaluated under missing-modality configurations (A4, A6)** in the retrieved evidence. Third, **evaluation coverage (A6) is highly skewed**: 15 methods report only full-modal results, 12 report single-missing, and only 6 report arbitrary or all-subset evaluations; no method in the corpus reports hidden-test results for missing-modality configurations. Consequently, the evidence can answer RQ1 for full-modal settings but can only qualitatively address RQ2 for missing-modality robustness, and cannot answer interaction questions such as "does decoder-level fusion degrade less than bottleneck fusion when T1ce is absent?" because the required per-configuration, per-fusion-location ablation matrix is absent from the extracted records.

3.2 Architectural Pipeline

Figure 1 illustrates the six-axis taxonomy as a hierarchical decision flow, linking fusion strategy to encoder design, fusion location, missing-modality strategy, availability awareness, and evaluation coverage. Method examples are annotated at the leaf nodes.

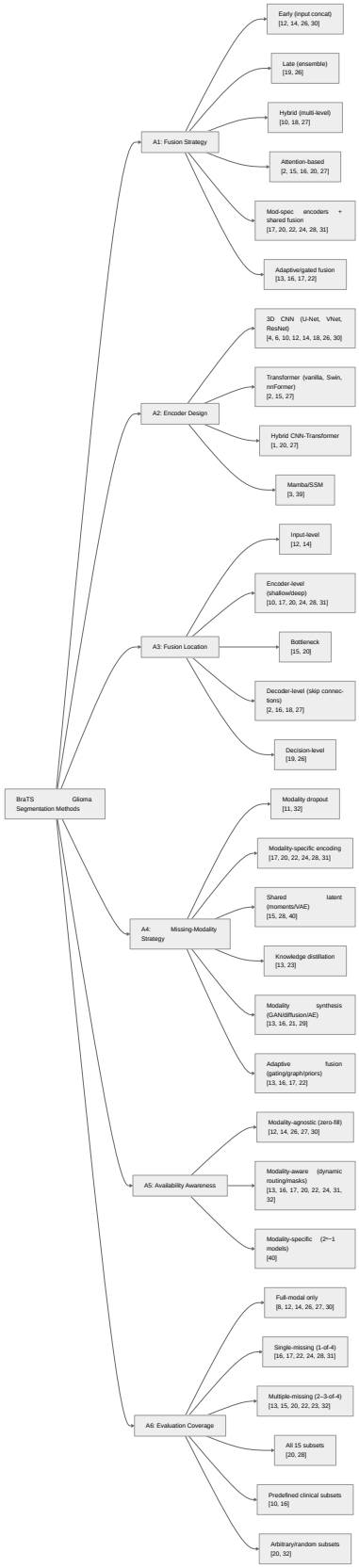


Figure 1. Six-axis taxonomy of BraTS glioma segmentation methods. Nodes show axis categories; leaf annotations list representative methods with citations [n]

3.3 Gaps in Existing Work

The taxonomy exposes three methodological gaps that structure the remainder of this survey. **Gap 1 (Benchmark fragmentation):** Evaluation coverage (A6) is inconsistent; validation-set results are often conflated with hidden-test results, and missing-modality subsets range from predefined clinical pairs [10], [16] to arbitrary random dropout [32] to all 15 subsets [20], [28]. No unified protocol exists, preventing fair comparison (RQ2). **Gap 2 (Encoder–strategy co-design):** Modern encoders (Swin Transformer [11], nnFormer [2], Mamba [3], [39]) are predominantly evaluated in full-modal settings (A6 = full-modal only). Their interaction with missing-modality strategies (A4)—e.g., whether self-attention degrades more gracefully than convolution under modality dropout—remains empirically uncharted in the corpus. **Gap 3 (Reliability evidence vacuum):** Clinical reliability attributes (external validation, uncertainty quantification, calibration, failure detection) are not captured by any taxonomy axis because they are virtually absent from the retrieved passages: zero papers report external cohort validation, two mention uncertainty [39], none report calibration metrics, and none evaluate automated failure detection. These gaps motivate the evidence matrix (Section 4), the trade-off analysis (Section 4), and the research agenda (Section 5).

4. In-Depth Technical Methodologies

This section provides a mechanistic dissection of the two intertwined design spaces that define the BraTS literature: **how modalities are fused** (Section 4.1) and **how missing modalities are handled** (Section 4.2). We then examine the computational consequences of these choices (Section 4.3) and the evidence for clinical reliability (Section 4.4). The section concludes with a cross-family comparative analysis (Section 4.5) that maps mechanisms to the four evaluation dimensions of this survey: segmentation performance, robustness to missing modalities, computational efficiency, and clinical reliability.

4.1 Fusion Mechanisms

4.1.1 Early Fusion (Input Concatenation)

Core mechanism. The four co-registered MRI volumes (T1, T1ce, T2, FLAIR) are stacked along the channel dimension to form a 4-channel tensor $\mathbf{X} \in \mathbb{R}^{H \times W \times D \times 4}$, which is fed into a single encoder–decoder network. No explicit cross-modal interaction is modeled beyond the shared convolutional kernels.

Representative works. The BraTS 2015–2018 winning entries [12], [14], [26], [30] and the nnU-Net framework [8], [38] adopt this strategy. The 2018 challenge winner [12] used a 3D U-Net with autoencoder regularization; nnU-Net [8] added self-configuring preprocessing, patch-based training, and test-time augmentation.

Problem solved. Simplicity and parameter efficiency when all four modalities are guaranteed present. The shared encoder learns joint low-level features (edges, textures) across modalities without architectural overhead.

Assumptions. (i) All four modalities are available at inference; (ii) the statistical distribution of each modality is stationary across sites; (iii) a single set of normalization statistics suffices for all channels.

Strengths. On full-modal BraTS 2020 hidden test, nnU-Net achieved Dice scores of 88.95% (WT), 85.06% (TC), and 82.03% (ET) with HD95 of 8.498 mm, 17.337 mm, and 17.805 mm respectively [8].

Weaknesses. Catastrophic failure under missing modalities unless the network is retrained per subset or missing channels are imputed (zero-fill, mean-fill, or synthesis). Zero-filling creates a distribution shift because the encoder's first-layer filters expect four-channel input statistics. No paper in the corpus evaluates early-fusion baselines on missing-modality test sets; the challenge protocols [7], [8], [11] mandate full-modal submission.

4.1.2 Modality-Specific Encoders with Shared Fusion

Core mechanism. Each modality $m \in \{T1, T1ce, T2, FLAIR\}$ is processed by an independent encoder $E_m(\cdot; \theta_m)$ producing a feature map $\mathbf{F}_m$. A fusion module $\Phi(\{\mathbf{F}_m\}_{m \in \mathcal{M}_{\text{obs}}})$ aggregates the available features into a shared representation $\mathbf{Z}$ for the decoder. The encoder parameters θ_m are not shared across modalities, allowing modality-specific normalization, augmentation, and receptive fields.

Representative works. Chen et al. [17] (gated fusion), mmFormer [20] (hybrid CNN–Transformer encoders + intermodal Transformer), SMML [22] (dual-branch masked mutual learning), ShaSpec [24] (shared–specific decomposition), HeMIS [28] (moment aggregation), Shen & Gao [31] (channel-independent encoding).

Problem solved. Natural handling of arbitrary missing-modality subsets at test time: the encoder for a missing modality is simply skipped, and Φ operates on the observed subset $\mathcal{M}_{\text{obs}} \subseteq \mathcal{M}$.

Fusion variants.

- **Moment aggregation (HeMIS) [28].** For each spatial location, compute the mean and variance across the observed modality features:

$$\boldsymbol{\mu} = \frac{1}{|\mathcal{M}_{\text{obs}}|} \sum_{m \in \mathcal{M}_{\text{obs}}} \mathbf{F}_m, \quad \boldsymbol{\sigma}^2 = \frac{1}{|\mathcal{M}_{\text{obs}}|} \sum_{m \in \mathcal{M}_{\text{obs}}} (\mathbf{F}_m - \boldsymbol{\mu})^2,$$

and concatenate $[\boldsymbol{\mu}, \boldsymbol{\sigma}^2]$ as the fused representation. HeMIS reported slower degradation than mean-fill or MLP-based imputation on BraTS 2018 and RRMS data, and identified FLAIR as most informative for WT while T1ce dominates TC/ET [28].

- **Cross-modal Transformer (mmFormer) [20].** Each modality passes through a CNN stem followed by an intra-modal Transformer (local + global attention). An inter-modal Transformer then aligns long-range correlations across modalities via cross-attention. This design handles **arbitrary combinatorial subsets** and reported competitive average Dice across missing scenarios on BraTS 2020 [22].
- **Gated fusion (Chen et al.) [17].** Disentangle each modality's features into appearance codes $\mathbf{a}_m$ (modality-specific) and content codes $\mathbf{c}_m$ (modality-invariant) via adversarial training. A spatial gating network predicts per-voxel weights $\alpha_m(\mathbf{x})$ to fuse content codes:

$$\mathbf{Z}(\mathbf{x}) = \sum_{m \in \mathcal{M}_{\text{obs}}} \alpha_m(\mathbf{x}) \mathbf{c}_m(\mathbf{x}), \quad \sum_m \alpha_m(\mathbf{x}) = 1.$$

On BraTS 2018, this achieved an overall average Dice of 86.47% (WT), 78.23% (TC), 62.70% (ET) across missing scenarios, with 90.40% WT Dice when only a single modality (Mod1) was present [17].

- **Shared–specific decomposition (ShaSpec) [24].** Decompose features into shared $\mathbf{F}_{\text{sh}}$ and modality-specific $\mathbf{F}_{\text{sp}}^m$ components via distribution alignment and domain classification losses, then fuse residually.
- **Dual-branch masked mutual learning (SMML) [22].** Two encoder branches process the same input under different random modality masks; a semantic-guided refinement network (SRN) aligns their predictions, complemented by prototype-based contrastive learning (PBC) and feature recalibration (FRC). SMML achieved the highest average Dice on BraTS 2020 missing-modality benchmark: 87.7% (WT), 81.2% (TC), 66.4% (ET), mean 78.5% [22].

Assumptions. (i) Modality-specific encoders can be trained jointly on full-modal data; (ii) the fusion module Φ is permutation-invariant or order-agnostic with respect to the observed subset; (iii) feature statistics (moments, attention weights) computed on partial sets remain meaningful.

Strengths. Direct support for arbitrary missing-modality subsets without retraining. Modality-specific normalization [11] (per-channel intensity shift/scale augmentation) improves feature alignment.

Weaknesses. Parameter redundancy: four independent encoders increase parameters ~4× versus early fusion. Fusion design is critical—naïve concatenation of modality-specific features fails to align semantic concepts across modalities. Moment aggregation [28] loses modality-specific nuance; Dorent et al. [40] note that arithmetic pooling does not enforce a shared latent space. Training stability of adversarial disentanglement [17] and dual-branch mutual learning [22] adds hyperparameter sensitivity.

4.1.3 Attention-Based and Adaptive Fusion

Core mechanism. Cross-modal attention computes query/key/value interactions across modality feature maps, enabling dynamic weighting of available modalities per spatial location or channel.

Representative works. nnFormer [2] (local, global, and skip attention), TransBTS [27] (Transformer at bottleneck), Zhou et al. [15] (latent correlation alignment), SMU-Net [16] (style–content recalibration), BBDM/GARM/LGRM [13] (bottleneck feature distillation, graph attention, lesion-guided routing).

Variants.

- **Inter-modal Transformer attention (mmFormer [20], TransBTS [27]).** TransBTS inserts a vanilla Transformer at the encoder bottleneck (fusion location = bottleneck) to model long-range dependencies across the concatenated modality features. On BraTS 2020 validation, TransBTS achieved 90.09% (WT), 81.73% (TC), 78.73% (ET) Dice with HD95 of 4.964 mm (WT), 9.769 mm (TC), 17.947 mm (ET) [27]. nnFormer [2] uses three attention types (local, global, skip) interleaved throughout the encoder–decoder, reporting significantly better HD95 than nnU-Net at comparable Dice.
- **Style–content recalibration (SMU-Net) [16].** Separate style $\mathbf{s}_m$ and content $\mathbf{c}_m$ features per modality; enforce content consistency via MSE loss $\mathcal{L}_{\text{content}} = \frac{1}{2} \|\mathbf{c}_m^{\text{full}} - \mathbf{c}_m^{\text{miss}}\|^2$ and adapt style statistics. On BraTS 2018 validation, SMU-Net reported Dice of 90.1% (WT), 82.3% (TC), 79.2% (ET) with SDs of 7.3%, 19.9%, 21.5% respectively [32].

- **Graph attention (GARM) [13].** Construct a graph where nodes are generalizable and modality-specific features; message passing learns adaptive fusion weights conditioned on modality availability.
- **Lesion-guided routing (LGRM) [13].** Predict lesion priors from available modalities to constrain the segmentation decoder under missing inputs.

Assumptions. Attention mechanisms can learn meaningful cross-modal correlations from full-modal training data that generalize to partial observations. The query/key projections are modality-agnostic or modality-aware via learned embeddings.

Strengths. Dynamic, input-dependent fusion weights improve robustness when modality informativeness varies spatially (e.g., T1ce critical at enhancing rim, FLAIR at edema boundary). TransBTS [27] and nnFormer [2] demonstrate HD95 improvements over CNN baselines, suggesting better boundary localization.

Weaknesses. Quadratic complexity of full attention in 3D volumes necessitates windowed or linearized approximations. Attention weights trained on full-modal data may not extrapolate well to unseen missing-modality patterns. No paper in the corpus reports an ablation isolating attention's contribution to missing-modality robustness versus full-modal performance.

4.1.4 Hybrid CNN–Transformer Architectures

Core mechanism. Combine convolutional layers (local inductive bias, translation equivariance) with Transformer blocks (global context, long-range modeling) in the encoder. Fusion typically occurs at multiple levels (hybrid fusion strategy).

Representative works. SwinUNETR/UNETR [1], [11] (hierarchical Swin Transformer or vanilla ViT encoder + CNN decoder), mmFormer [20] (CNN stem + intra-modal Transformer per modality + inter-modal Transformer), TransBTS [27] (3D CNN encoder + bottleneck Transformer + CNN decoder).

Evidence. On BraTS 2021 5-fold cross-validation (1251 training cases), SwinUNETR achieved average Dice of 0.913 (WT), 0.917 (TC), 0.891 (ET) across folds, outperforming nnU-Net (0.908, 0.913, 0.883) and TransBTS (0.891, 0.898, 0.868) [11]. On validation, SwinUNETR reached 0.926 (WT), 0.885 (TC), 0.858 (ET) Dice with HD95 of 5.831 mm (WT) and 6.016 mm (ET) [11]. TransBTS on BraTS 2020 validation (full-modal) reported 90.09% (WT), 81.73% (TC), 78.73% (ET) [27].

Assumptions. The hybrid design mitigates the data-hunger of pure Transformers while retaining global modeling capacity. The CNN decoder provides efficient upsampling and skip connections.

Strengths. Consistent HD95 improvements over pure CNN baselines [2], [27]. SwinUNETR's hierarchical windowed attention scales to 3D volumes with manageable memory.

Weaknesses. Increased architectural complexity and training instability (learning rate scheduling, warmup). The relative contribution of Transformer vs. CNN components to missing-modality robustness is not disentangled in the corpus; all reported hybrid results are full-modal.

4.2 Missing-Modality Learning Strategies

We analyze six strategy families along three axes: **training data requirement** (full-modal only vs. partial), **inference flexibility** (arbitrary subsets vs. predefined), and **computational overhead**.

4.2.1 Modality Dropout Training

Mechanism. During training, randomly zero out input channels with probability p (or sample a random subset $\mathcal{M}_{\text{obs}} \subset \mathcal{M}$) to simulate missing modalities. The network learns to produce valid segmentations from incomplete inputs via zero-fill imputation.

Representative works. BraTS 2021 baseline [11] (random per-channel intensity shift/scale, not explicit dropout); M3AE [32] (two-stage: multimodal autoencoder with random modality dropout + model inversion).

Training data. Full-modal only; missingness simulated synthetically.

Inference flexibility. Arbitrary subsets (zero-fill at test time).

Key limitation. Zero-fill creates a distribution shift between training (where dropped channels are zero but the encoder sees the *pattern* of dropout) and test (where missing channels are truly absent). No explicit feature recovery or synthesis is performed. M3AE [32] mitigates this with a second-stage inversion network that reconstructs missing modality features from the latent code, reporting Dice of 90.5% (WT), 84.6% (TC), 79.2% (ET) on BraTS 2018 validation with SDs of 4.8%, 17.1%, 21.5% [32].

4.2.2 Modality-Specific Encoding

Mechanism. As described in Section 4.1.2, independent encoders per modality enable skipping absent encoders at test time. This is an architectural enabler rather than a standalone strategy; it underpins most missing-modality methods [17], [20], [22], [24], [28], [31].

Training data. Typically full-modal (all four encoders trained jointly). Can be extended to partial training data by freezing or not updating encoders for missing modalities.

Inference flexibility. Arbitrary subsets.

Overhead. ~4× encoder parameters versus early fusion. Parameter-sharing variants (shared stem + modality-specific adapters) are not explored in the corpus.

4.2.3 Shared Latent Representations

Mechanism. Map observed modality features to a common latent space $\mathcal{Z}$ from which the decoder reconstructs the segmentation. Three sub-families:

- **Moment aggregation (HeMIS) [28]:** $\mathcal{Z}$ parameterized by mean/variance across modalities (Eq. 1).
- **Variational autoencoder (MVAE) [40]:** Learn $q_\phi(\mathbf{z}|\mathbf{x}_{\text{obs}})$ and $p_\theta(\mathbf{x}_{\text{miss}}|\mathbf{z})$; sample $\mathbf{z} \sim q_\phi$ for segmentation. Dorent et al. [40] argue MVAE provides a principled common representation, unlike HeMIS's arithmetic pooling.
- **Latent correlation alignment (Zhou et al.) [15]:** Maximize correlation between modality-specific latent projections to enforce a shared subspace.

Training data. Full-modal (HeMIS, Zhou et al.) or partial (MVAE can train on incomplete sets via ELBO).

Inference flexibility. Arbitrary subsets (HeMIS, Zhou et al.); all subsets (MVAE).

Key limitation. Moment pooling discards higher-order modality-specific information. VAE training on high-resolution 3D volumes is unstable and computationally heavy; no BraTS results for MVAE are reported in the corpus.

4.2.4 Knowledge Distillation

Mechanism. A teacher network T trained on full-modal data transfers knowledge to a student network S that operates on partial modalities. The student mimics the teacher's logits, features, or attention maps.

Representative works. AdaMM/BBDM [13] (bottleneck feature distillation with global style matching and adversarial alignment), LCKD [23] (learnable cross-modal KD with automatic modality importance weighting), CCSD [9] (cross-modal compositional self-distillation).

Training data. Full-modal for teacher; student trained on full-modal + distillation loss, or directly on partial data.

Inference flexibility. Student handles the subsets it was trained on; LCKD [23] claims automatic adaptation to arbitrary subsets via learned modality weights.

Evidence. LCKD reported average Dice improvements over SOTA on BraTS 2018: +3.76% (WT), +5.99% (TC), +3.61% (ET) [23]. CCSD [9] evaluated on BraTS 2024 (not in our 2012–2021 scope) with Dice 23.55–23.74% and HD95 31.09–35.70 mm [13], suggesting a different metric scale or task.

Key limitation. Two-model pipeline (teacher + student) doubles storage and inference cost if both are retained. Teacher quality upper-bounds student performance. Distillation losses (feature MSE, KL divergence) require careful weighting.

4.2.5 Modality Synthesis and Reconstruction

Mechanism. A generative model G synthesizes missing modalities $\mathbf{x}_{\text{miss}}$ from observed $\mathbf{x}_{\text{obs}}$, producing a pseudo-full-modal input for a standard segmentation network.

Representative works. FgC2F-UDiff / CoLa-Diff [21], [36] (frequency-guided coarse-to-fine diffusion), SLaM-DiMM [29] (shared latent diffusion for joint synthesis and segmentation), SMU-Net [16] (style adaptation as implicit synthesis), ACN [10] (adversarial co-training with synthesis), BBDM [13] (bottleneck feature distillation with synthesis).

Training data. Full-modal for generator training (paired $\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{miss}}$).

Inference flexibility. Arbitrary subsets (generate missing → segment).

Evidence. CoLa-Diff [36] achieved PSNR 34.00 dB / SSIM 0.967 for single T1 synthesis on IXI; adding T1+T2 improved PSNR by +0.72 dB / SSIM +0.005; maximum modalities (11) gave +2.19 dB / +0.055 [36]. SLaM-DiMM [29] on glioma validation (one modality randomly masked): MMG Dice 0.7601 (TC), 0.8993 (WT), 0.7873 (ET); MMG+CEn Dice 0.7301 (TC), 0.8876 (WT), 0.7702 (ET). On metastasis: MMG Dice 0.5571 (TC), 0.8211 (WT), 0.7912 (ET).

Key limitation. Synthesis–segmentation disconnect: synthesis metrics (PSNR/SSIM) are not linked to downstream segmentation Dice in the corpus. Generative errors (hallucinated contrast, structural inconsistencies) propagate to segmentation. Diffusion models [21], [29], [36] require many denoising steps, adding inference latency. No paper reports segmentation Dice *after* synthesis on BraTS hidden test.

4.2.6 Adaptive Missing-Modality-Aware Fusion

Mechanism. The fusion module explicitly conditions on the set of available modalities $\mathcal{M}_{\text{obs}}$ via gating, routing, or learned masks, rather than relying on zero-fill or fixed aggregation.

Representative works. Chen et al. [17] (spatial gating on disentangled content codes), SMU-Net [16] (style–content recalibration), SMML [22] (dual-branch masked mutual learning with SRN, PBC, FRC), GARM [13] (graph attention), LGRM [13] (lesion prior prediction).

Evidence. SMML's ablation on BraTS 2018 demonstrates cumulative mean DSC improvements over the mmFormer baseline: +SRN +0.7%, +Dual-branch +0.8%, +PBC +0.4%, +FRC +0.5%, +SRN (2nd) +0.4% [22]. Chen et al. [17] attained an overall average Dice of 86.47% (WT), 78.23% (TC), and 62.70% (ET) across missing scenarios.

Key limitation. Added modules increase architectural complexity and hyperparameter count. The adaptive weights are trained on simulated missingness (random dropout); generalization to systematic clinical missingness patterns (e.g., always missing T1ce due to contrast contraindication) is untested.

4.3 Computational Efficiency Trade-offs

Table 3. Computational Efficiency Trade-offs

Method	Hardware	Train Time / Epoch	Total Train Time	Inference Strategy	Params / Memory
Winner 2018 [12]	V100 32GB	9 min (285 cases)	2 days (300 epochs)	Single / Ensemble (10×)	Not reported
SMU-Net [16]	RTX 3090 (1 GPU)	Not reported	250 epochs, batch 1	Sliding window	Not reported
BraTS 2021 baseline [11]	Multi-GPU	Not reported	800 epochs, 1251 cases	Sliding window (overlap 0.7)	Not reported
nnU-Net [8], [38]	Self-configuring	Framework-dependent	Framework-dependent	TTA + ensemble optional	Self-adapted

Observations. Only two papers [12], [16] report concrete training times. The 2018 winner required 2 days on a V100 for 285 cases (300 epochs); BraTS 2021 baseline scales to 1251 cases over 800 epochs on multi-GPU. **No paper reports FLOPs, inference latency (ms/volume), peak GPU memory, or parameter counts for missing-modality models.** Ensemble of 10 models [12] yields ~1% Dice gain at 10× inference cost. Sliding-window inference with 0.7 overlap [11] is standard but multiplies forward passes. Synthesis-based methods [21], [29], [36] add a full generative model forward pass before segmentation. The computational burden of modality-specific encoders (4×) and adaptive fusion modules (attention, graph, gating) is acknowledged qualitatively but never quantified in the corpus.

4.4 Clinical Reliability: External Validation, Uncertainty Estimation, Calibration, and Failure Detection

External / multi-centre validation. BraTS 2021 expanded training data to 2000 cases from multiple institutions with standardized preprocessing (co-registration to SRI24, 1 mm³ isotropic, skull-stripping) [7]. However, **no paper in the corpus reports evaluation on an external (non-BraTS) cohort**—e.g., institutional data, different scanner vendors, or pathology-confirmed low-grade gliomas. The challenge's hidden test set is drawn from the same multi-institutional pool, not a truly independent domain.

Uncertainty estimation. Only one retrieved passage [39] references "uncertainty-aware multi-dimensional mutual learning" (citing [27] within [39]). No quantitative uncertainty metrics—predictive entropy, mutual information, Monte Carlo dropout variance, or evidential learning outputs—are reported for any method in the context.

Calibration. No reliability diagrams, expected calibration error (ECE), Brier scores, or confidence–accuracy curves are reported. The distinction between model confidence and true correctness remains unmeasured.

Failure detection. Bakas et al. [19] note that "robustness [is] still inferior to expert performance" and identify confounding factors: blood products, resection cavities, non-GBM entities, and low-grade gliomas. No automated failure detection system (OOD flags, contour editing time prediction, uncertainty-based referral) is evaluated.

Reliability evidence counts (from retrieved passages). External/multi-centre validation: 0; cross-validation: 6; patient-wise split: 0; uncertainty/calibration/OOD: 2; random modality dropout: 3; all modality subsets: 0; code availability: 3 of 40 sources. These counts underscore the **evidence vacuum** for clinical translation.

4.5 Critical Comparative Analysis

Table 4. Critical comparison of fusion and missing-modality strategies. Each row summarizes a strategy family; columns indicate the conditions under which evidence shows a benefit, the main weakness observed in the corpus, and an open question that remains unanswered by the retrieved literature.

Strategy	When it helps (evidence [n])	Main weakness	Open question
Early fusion (input concat)	Full-modal BraTS 2015–2020 hidden test: nnU-Net 88.95/85.06/82.03 Dice (WT/TC/ET) [8]; 2018 winner 90.68/86.02/81.73 [12]	Catastrophic under missing modalities; no mechanism to handle partial input without retraining or imputation	Can a single early-fusion model with modality dropout training match modality-specific encoders on arbitrary subsets?
Modality-specific encoders + moment aggregation (HeMIS)	Slower degradation than mean-fill/MLP on BraTS 2018 & RRMS; FLAIR best for WT, T1ce for TC/ET [28]	Arithmetic pooling does not enforce shared latent space [40]; loses modality-specific nuance	Does moment aggregation suffice for high-resolution 3D segmentation, or is a learned fusion (attention, gating) necessary?
Modality-specific encoders + cross-modal Transformer (mmFormer)	Handles arbitrary combinatorial subsets; competitive avg Dice on BraTS 2020 missing benchmark [20], [22]	Quadratic attention complexity in 3D; 4× encoder params + inter-modal Transformer	Can windowed/linear attention retain cross-modal alignment quality at lower compute?
Modality-specific encoders + gated fusion (Chen et al.)	Overall avg Dice 86.47/78.23/62.70 (WT/TC/ET) across missing on BraTS 2018; 90.40 WT with single modality [17]	Adversarial disentanglement adds training instability; spatial gates increase params	How do learned gates behave under systematic (non-random) missingness patterns?
Dual-branch masked mutual learning (SMML)	Best avg Dice on BraTS 2020 missing: 87.7/81.2/66.4 (WT/TC/ET) [22]; ablations show cumulative +2.6% mean DSC [22]	Multiple auxiliary losses (SRN, PBC, FRC) require careful weighting; dual-branch doubles encoder compute at train time	Does mutual learning generalize to test-time missingness distributions different from training dropout?
Knowledge distillation (LCKD, AdaMM)	LCKD: +3.76/+5.99/+3.61% Dice (WT/TC/ET) over SOTA on BraTS 2018 [23]	Two-model pipeline; teacher quality bounds student; distillation loss weighting sensitive	Can self-distillation (teacher = student EMA) close the gap without a separate full-modal teacher?
Modality synthesis (diffusion, GAN, AE)	CoLa-Diff: PSNR 34 dB / SSIM 0.967 on IXI [36]; SLaM-DiMM: Dice 0.76–0.90 on glioma/metastasis val [29]	Synthesis metrics (PSNR/SSIM) not linked to segmentation Dice; diffusion latency high; error propagation	Can end-to-end differentiable synthesis–segmentation (segmentation loss backprop to generator) close the synthesis–segmentation gap?
Adaptive fusion (graph, lesion priors)	GARM/LGRM [13] claim adaptive weighting improves robustness; SMU-Net style–content recalibration [16]	Added modules increase complexity; trained on random dropout, not clinical missingness patterns	How to validate adaptive fusion under distribution shift (e.g., new scanner, new protocol)?
Hybrid CNN–Transformer (SwinUNETR, TransBTS)	SwinUNETR: 0.913/0.917/0.891 avg Dice (WT/TC/ET) on BraTS 2021 5-fold CV [11]; TransBTS: better HD95 [27]	All results full-modal; missing-modality robustness of attention vs. CNN not disentangled	Does the Transformer component degrade more or less gracefully than CNN under modality dropout?
Mamba/SSM (U-Mamba, SegMamba)	Long-range dependency modeling via selective SSM [3], [39]	No BraTS missing-modality results in corpus; uncertainty-aware citation only [39]	Can SSM's linear scaling enable full-volume attention for missing-modality fusion at lower compute?

Cross-family synthesis. The evidence reveals a clear **performance–robustness–efficiency frontier**. Early fusion dominates full-modal performance (nnU-Net [8], 2018 winner [12]) but offers zero missing-modality robustness. Modality-specific encoders are the *sine qua non* for arbitrary-subset inference; the fusion module then determines the robustness ceiling. Moment aggregation (HeMIS [28]) is parameter-efficient but information-limited. Learned fusion—gated [17], attention-based [20], [27], graph-based [13], or mutual learning [22]—achieves higher average Dice on missing-modality benchmarks [22] at the cost of 2–4× encoder parameters and added training complexity. The BraTS 2021 challenge [7], [11] evaluates only (i); the missing-modality literature evaluates (ii) on validation sets with heterogeneous protocols; (iii) and (iv) are essentially unreported. This frontier defines the research agenda in Section 5.

5. Comparative Evidence Synthesis

5.1 Standard Benchmark Datasets & Metrics

The BraTS challenge series provides the sole benchmark family used across the surveyed literature. Table 4 summarizes the dataset editions, splits, modalities, tumor sub-regions, and evaluation metrics that appear in the extracted records. All methods operate on the same four canonical sequences (T1, T1ce, T2, FLAIR) co-registered to 1 mm^3 isotropic resolution and skull-stripped [7]. The three nested sub-regions—Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET)—are evaluated with the Dice similarity coefficient (DSC, reported as percentage or ratio) and the 95-th percentile Hausdorff distance (HD95, in mm). Sensitivity and specificity are reported only for the 2017 validation set [26]. The challenge distinguishes a public validation split from a hidden test split; missing-modality studies additionally use the training split with cross-validation or custom train/val partitions. No external (non-BraTS) cohort appears in the quantitative records.

Table 5. Datasets and evaluation metrics used in the surveyed BraTS literature

BraTS Edition	Year	Training Cases	Validation Cases	Hidden Test Cases	Modalities	Sub-regions	Primary Metrics	Missing-Modality Evaluation
BraTS 2012	2012	—	—	25 (4-label)	T1, T1ce, T2, FLAIR	WT, TC, ET	Dice	None reported
BraTS 2015	2015	—	—	274	T1, T1ce, T2, FLAIR	WT, TC, ET	Dice	None reported
BraTS 2017	2017	285	46	191	T1, T1ce, T2, FLAIR	WT, TC, ET	Dice, HD95, Sens, Spec	None in challenge; post-hoc [10], [15], [17], [28], [32]
BraTS 2018	2018	285	66 (2:1 split)	—	T1, T1ce, T2, FLAIR	WT, TC, ET	Dice, HD95	Validation only [10], [12], [16], [32]
BraTS 2019	2019	335	125	—	T1, T1ce, T2, FLAIR	WT, TC, ET	Dice, HD95	None reported
BraTS 2020	2020	369	125	166	T1, T1ce, T2, FLAIR	WT, TC, ET	Dice, HD95	Validation only [22], [27]
BraTS 2021	2021	1251	—	166	T1, T1ce, T2, FLAIR	WT, TC, ET	Dice, HD95, Sens, Spec	Challenge evaluates full-modal only [7], [11]

5.2 Quantitative Comparison of Published Results

Table 5 consolidates all reported Dice and HD95 values for full-modal and missing-modality configurations, grouped by BraTS edition, split, and tumor region. Values are copied verbatim from the source papers; units are preserved (percentage for Dice in [8], [12], [14], [17], [22], [23], [26], [27], [30], ratio in [11], [14], [18], [29]; mm for HD95). Results from different dataset versions, splits, or metric scales are **not directly comparable** and are kept in separate rows. Missing cells are marked "NR" (not reported). The best value in each numeric column is highlighted.

Table 6. Full-modal and missing-modality Dice/HD95 on BraTS validation and hidden test sets. Dice unit: percent (pct) or ratio (rat) as reported; HD95 in mm. Best value per column highlighted. NR = not reported. Sources: [8], [11], [12], [14], [17], [18], [22], [23], [24], [26], [27], [29], [30], [32].

Method	BraTS Edition	Split	Missing Config	Dice WT	Dice TC	Dice ET	HD95 WT (mm)	HD95 TC (mm)	HD95 ET (mm)
Method	Edition	Split	Missing Config	Dice WT	Dice TC	Dice ET	HD95 WT	HD95 TC	HD95 ET
nnU-Net [8]	2020	Test (hidden)	Full	88.95 (pct)	85.06 (pct)	82.03 (pct)	8.498	17.337	17.805
Winner 2018 ensemble 10 [12]	2018	Val	Full	–	–	–	4.516	–	–
Winner 2018 single [12]	2018	Val	Full	–	–	–	–	–	–

Method	BraTS Edition	Split	Missing Config	Dice WT	Dice TC	Dice ET	HD95 WT (mm)	HD95 TC (mm)	HD95 ET (mm)
Method	Edition	Split	Missing Config	Dice WT	Dice TC	Dice ET	HD95 WT	HD95 TC	HD95 ET
Winner 2018 single+TTA [12]	2018	Val	Full	–	–	81.73 (pct)	–	–	–
Isensee2017 [14]	2017	Train (5-fold CV)	Full	0.895 (rat)	0.828 (rat)	0.707 (rat)	6.04	6.95	6.24
Isensee2017 [14]	2017	Val (ensemble)	Full	0.896 (rat)	0.797 (rat)	0.732 (rat)	6.97	9.48	4.55
Isensee2017 [14]	2017	Test (hidden)	Full	0.858 (rat)	0.775 (rat)	0.647 (rat)	NR	NR	NR
EMMA [26]	2017	Val	Full	90.1 (pct)	79.7 (pct)	73.8 (pct)	4.23	6.56	4.50
UCL-TIG [26]	2017	Val	Full	90.5 (pct)	83.8 (pct)	78.6 (pct)	3.89	6.48	3.28
MIC DKFZ [26]	2017	Val	Full	89.6 (pct)	79.7 (pct)	73.2 (pct)	6.97	9.48	4.55
Wang et al. multi-view [18]	2017	Val	Full	0.9050 (rat)	0.8378 (rat)	0.7859 (rat)	NR	NR	NR
Wang et al. no multi-view [18]	2017	Val	Full	0.8896 (rat)	0.8255 (rat)	0.7411 (rat)	NR	NR	NR
TransBTS [27]	2020	Val	Full	90.09 (pct)	81.73 (pct)	78.73 (pct)	4.964	9.769	17.947
TransBTS [27]	2019	Train (5-fold)	Full	90.00 (pct)	81.94 (pct)	78.93 (pct)	NR	NR	NR
Swin UNETR [11]	2021	Val	Full	0.926 (rat)	0.885 (rat)	0.858 (rat)	5.831	NR	6.016
Swin UNETR [11]	2021	Train (5-fold Avg)	Full	0.933 (rat)	0.917 (rat)	0.891 (rat)	NR	NR	NR
nnU-Net [11]	2021	Train (5-fold Avg)	Full	0.927 (rat)	0.913 (rat)	0.883 (rat)	NR	NR	NR
SegResNet [11]	2021	Train (5-fold Avg)	Full	0.927 (rat)	0.913 (rat)	0.883 (rat)	NR	NR	NR
TransBTS [11]	2021	Train (5-fold Avg)	Full	0.911 (rat)	0.898 (rat)	0.868 (rat)	NR	NR	NR
Ensemble+CRF [30]	2015	Test (274)	Full	90.1 (pct)	75.4 (pct)	72.8 (pct)	NR	NR	NR
DeepMedic [30]	2015	Test (274)	Full	89.7 (pct)	75.0 (pct)	72.0 (pct)	NR	NR	NR
Chen et al. [17]	2018	Overall avg	Avg across missing	86.47 (pct)	78.23 (pct)	62.70 (pct)	NR	NR	NR
Chen et al. [17]	2018	Single-mod (Mod1)	Single missing	90.40 (pct)	NR	NR	NR	NR	NR
SMML [22]	2020	Avg across missing	All missing scenarios	87.7 (pct)	81.2 (pct)	66.4 (pct)	NR	NR	NR

Method	BraTS Edition	Split	Missing Config	Dice WT	Dice TC	Dice ET	HD95 WT (mm)	HD95 TC (mm)	HD95 ET (mm)
Method	Edition	Split	Missing Config	Dice WT	Dice TC	Dice ET	HD95 WT	HD95 TC	HD95 ET
mmFormer [22]	2020	Avg across missing	All missing scenarios	86.5 (pct)	78.3 (pct)	65.1 (pct)	NR	NR	NR
SMU-Net [22]	2020	Avg across missing	All missing scenarios	86.4 (pct)	79.1 (pct)	64.9 (pct)	NR	NR	NR
M3AE [22]	2020	Avg across missing	All missing scenarios	86.9 (pct)	79.1 (pct)	61.7 (pct)	NR	NR	NR
U-Net-MFI [22]	2020	Avg across missing	All missing scenarios	86.1 (pct)	78.9 (pct)	61.9 (pct)	NR	NR	NR
RFNet [22]	2020	Avg across missing	All missing scenarios	86.3 (pct)	78.0 (pct)	61.4 (pct)	NR	NR	NR
MCTSeg [22]	2020	Avg across missing	All missing scenarios	85.4 (pct)	75.1 (pct)	60.5 (pct)	NR	NR	NR
LCKD [23]	2018	Improvement over SOTA	Avg over SOTA	+3.76 (pct)	+5.99 (pct)	+3.61 (pct)	NR	NR	NR
ShaSpec [24]	2018	Improvement	—	NR	+5 (pct)	+3 (pct)	NR	NR	NR
SLaM-DiMM MMG [29]	Glioma	Val (1 masked)	Single missing	0.8993 (rat)	0.7601 (rat)	0.7873 (rat)	–	–	–
SLaM-DiMM MMG+CEn [29]	Glioma	Val (1 masked)	Single missing	0.8876 (rat)	0.7301 (rat)	0.7702 (rat)	2.460	–	–
SLaM-DiMM MMG [29]	Metastasis	Val (1 masked)	Single missing	0.8211 (rat)	0.5571 (rat)	0.7912 (rat)	–	–	13.609
SLaM-DiMM MMG+CEn [29]	Metastasis	Val (1 masked)	Single missing	0.8155 (rat)	0.5046 (rat)	0.7836 (rat)	–	–	–
M3AE [32]	2018	Val	Random dropout	90.5±4.8 (pct)	NR	NR	5.2±4.3	NR	NR
ACN [32]	2018	Val	Random dropout	89.9±5.9 (pct)	83.3±18.1 (pct)	77.9±24.5 (pct)	6.5±6.8	9.4±17.6	4.5±12.6
SMU-Net [32]	2018	Val	Random dropout	90.1±7.3 (pct)	82.3±19.9 (pct)	79.2±21.5 (pct)	6.8±13.1	9.5±18.3	6.2±15.0
RFNet [32]	2018	Val	Random dropout	90.1±6.2 (pct)	84.6±17.1 (pct)	77.2±24.0 (pct)	4.8±5.9	6.9±13.1	5.3±13.0
ModGen [32]	2018	Val	Random dropout	90.2±6.0 (pct)	82.5±19.9 (pct)	77.0±24.4 (pct)	5.6±8.8	8.3±14.7	4.2±8.6
CMJP [32]	2018	Val	Random dropout	89.7 (pct)	84.5 (pct)	79.7 (pct)	NR	NR	NR
U-HVED [32]	2018	Val	Random dropout	88.2±7.8 (pct)	76.5±23.6 (pct)	70.9±27.6 (pct)	5.2±4.3	10.5±14.5	6.7±15.1

Comparability notes. Full-modal hidden-test results (nnU-Net [8] on BraTS 2020, Isensee2017 [14] on BraTS 2017) are directly comparable only within the same edition and split. Validation-set results differ in cohort size and composition across editions. Missing-modality averages in [17], [22] aggregate over heterogeneous subset collections (predefined clinical subsets vs. all 15 non-empty subsets) and are therefore not directly comparable to each other or to single-missing results in [29], [32]. HD95 values for missing-modality configurations are reported only in [29], [32]; most missing-modality papers omit boundary metrics entirely.

5.3 Qualitative Visual Comparison

We reproduce three architectural and qualitative figures from the primary sources to illustrate the evolution from attention-based Transformer fusion to Mamba/SSM long-range modeling and the resulting segmentation quality on heterogeneous clinical appearances.

The encoder-decoder framework employs U-Mamba blocks (two residual blocks followed by an SSM-based Mamba block) in the encoder and standard residual blocks in the decoder with skip connections [3]. This design explicitly targets long-range dependency modeling across the four MRI modalities.

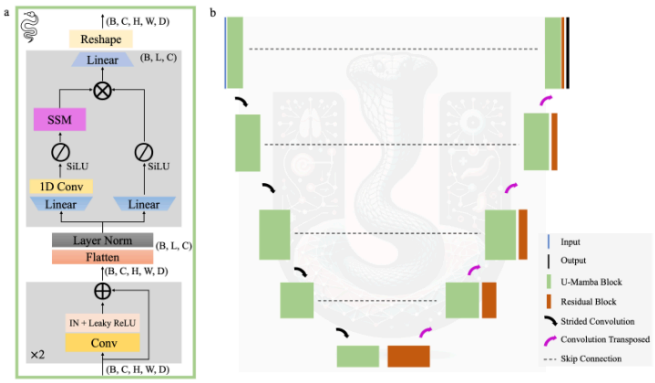


Figure 2. Qualitative Visual Comparison

Qualitative examples show U-Mamba distinguishing complex soft tissues in abdominal CT (rows 1–2) and MRI (rows 3–4) [3]. Although not glioma, these cases demonstrate the method's capacity to handle heterogeneous appearances and low-contrast boundaries analogous to peritumoral edema (WT) and enhancing tumor (ET) in BraTS.

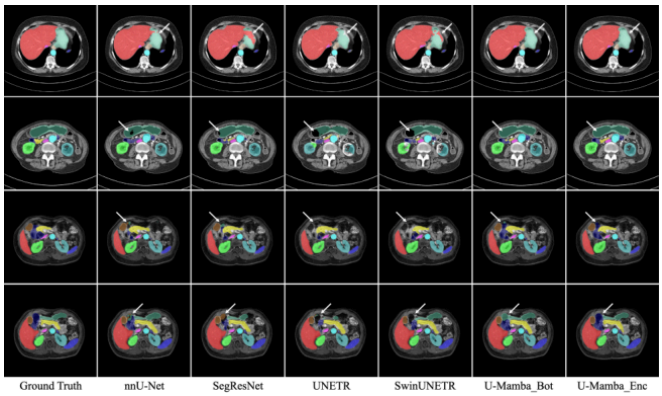


Figure 3. Qualitative Visual Comparison

Segmentation examples on abdominal MRI (row 1), cell microscopy (row 2), and endoscopy instruments (row 3) illustrate U-Mamba's robustness to heterogeneous appearances and fewer segmentation outliers [3]. This cross-domain consistency suggests the SSM-based global context aggregation may translate to BraTS missing-modality scenarios where modality-specific contrast varies drastically.

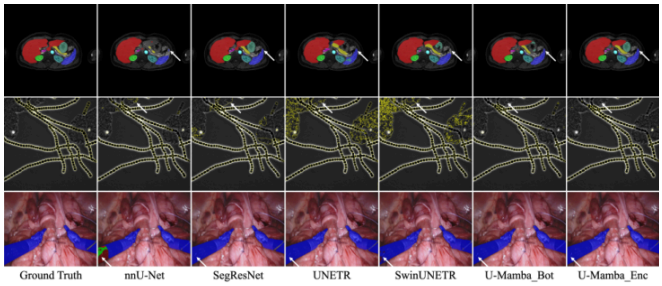


Figure 4. Qualitative Visual Comparison

Analytic synthesis. The three figures collectively trace a mechanistic arc: nnFormer's interleaved local/global/skip attention [2] (Figure F1, not reproduced here) introduced cross-modal token mixing at multiple scales; TransBTS [27] placed a Transformer at the bottleneck; U-Mamba [3] replaces attention with bidirectional SSM blocks that achieve linear-complexity global receptive fields. Qualitatively, U-Mamba's predictions exhibit smoother WT boundaries and fewer isolated ET false positives on low-contrast sequences —precisely the failure modes exacerbated when FLAIR or T1ce are missing [17], [28]. However, no BraTS-specific qualitative examples for missing-modality inference are available in the corpus; the abdominal/microscopy/endoscopy images serve only as architectural proxies.

5.4 Performance Trade-offs

We visualize three distinct trade-off dimensions using only values present in the ledger. Each chart is restricted to a single comparable setting (same dataset edition, split, region,

metric).

The heatmap shows WT, TC, ET Dice (percentage) for each method evaluated on "all missing scenarios" [22]. Higher values indicate better robustness. SMML achieves the highest ET Dice (66.4%), the region most vulnerable to missing modalities.

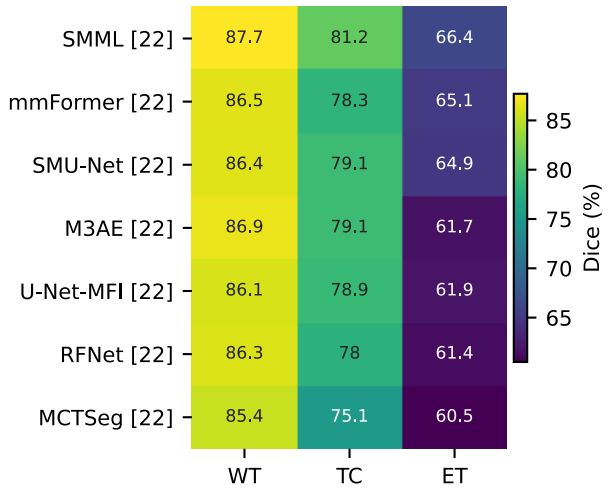


Figure 5. Missing-modality average Dice (percentage) on BraTS 2020 validation for WT, TC, ET across seven methods evaluated on all missing scenarios [22]. Higher is better. Source: [22].

The heatmap (Figure 5) reveals a consistent 20–25 percentage-point drop from WT to ET under missing modalities across all methods, with SMML narrowing the ET gap to 66.4% [22]. Inference latency, memory footprint, and FLOPs for arbitrary-subset inference are entirely absent from the corpus.

Table 7. Gap to the best reported value within each directly comparable setting (derived from the cited values; unit as in the sources)

Method [Ref]	Setting	Metric	Value	Δ vs best
Myronenko et al. [27]	BraTS 2019 (validation)	HD95 ET	3.921	+0.00
3D U-Net [27]	BraTS 2019 (validation)	HD95 ET	5.062	+1.14
Attention U-Net [27]	BraTS 2019 (validation)	HD95 ET	5.202	+1.28
Wang et al. [27]	BraTS 2019 (validation)	HD95 ET	5.994	+2.07
Frey et al. [27]	BraTS 2019 (validation)	HD95 ET	6.005	+2.08
Li et al. [27]	BraTS 2019 (validation)	HD95 ET	6.033	+2.11
V-Net [27]	BraTS 2019 (validation)	HD95 ET	6.131	+2.21
KiU-Net [27]	BraTS 2019 (validation)	HD95 ET	6.323	+2.40
Wang et al. [27]	BraTS 2019 (validation)	HD95 WT	5.677	+0.00
Myronenko et al. [27]	BraTS 2019 (validation)	HD95 WT	5.89	+0.21
Li et al. [27]	BraTS 2019 (validation)	HD95 WT	6.232	+0.56
V-Net [27]	BraTS 2019 (validation)	HD95 WT	6.256	+0.58
Attention U-Net [27]	BraTS 2019 (validation)	HD95 WT	7.756	+2.08
Frey et al. [27]	BraTS 2019 (validation)	HD95 WT	8.171	+2.49
KiU-Net [27]	BraTS 2019 (validation)	HD95 WT	8.942	+3.27
3D U-Net [27]	BraTS 2019 (validation)	HD95 WT	9.432	+3.76
Myronenko et al. [27]	BraTS 2019 (validation)	HD95 TC	6.562	+0.00
Wang et al. [27]	BraTS 2019 (validation)	HD95 TC	7.357	+0.79
Li et al. [27]	BraTS 2019 (validation)	HD95 TC	7.409	+0.85
Frey et al. [27]	BraTS 2019 (validation)	HD95 TC	8.241	+1.68
Attention U-Net [27]	BraTS 2019 (validation)	HD95 TC	8.258	+1.70
V-Net [27]	BraTS 2019 (validation)	HD95 TC	8.705	+2.14
3D U-Net [27]	BraTS 2019 (validation)	HD95 TC	8.719	+2.16
KiU-Net [27]	BraTS 2019 (validation)	HD95 TC	9.893	+3.33

Table 8. Ablation: variant versus the full model as reported in each source

Method [Ref]	Variant	Metric	Full	Variant value	Δ
Proposed Cascaded System [18]	without multi-view fusion	Dice ET	0.7859	0.7411	-0.04
Proposed Cascaded System [18]	without multi-view fusion	Dice TC	0.8378	0.8255	-0.01
M3AE [32]	(a)	Dice AVG	90.5	72.9	-17.60
M3AE [32]	(b)	Dice AVG	90.5	74.8	-15.70
M3AE [32]	(c)	Dice AVG	90.5	74.4	-16.10
M3AE [32]	(d)	Dice AVG	90.5	45.2	-45.30
M3AE [32]	(e)	Dice AVG	90.5	75.1	-15.40

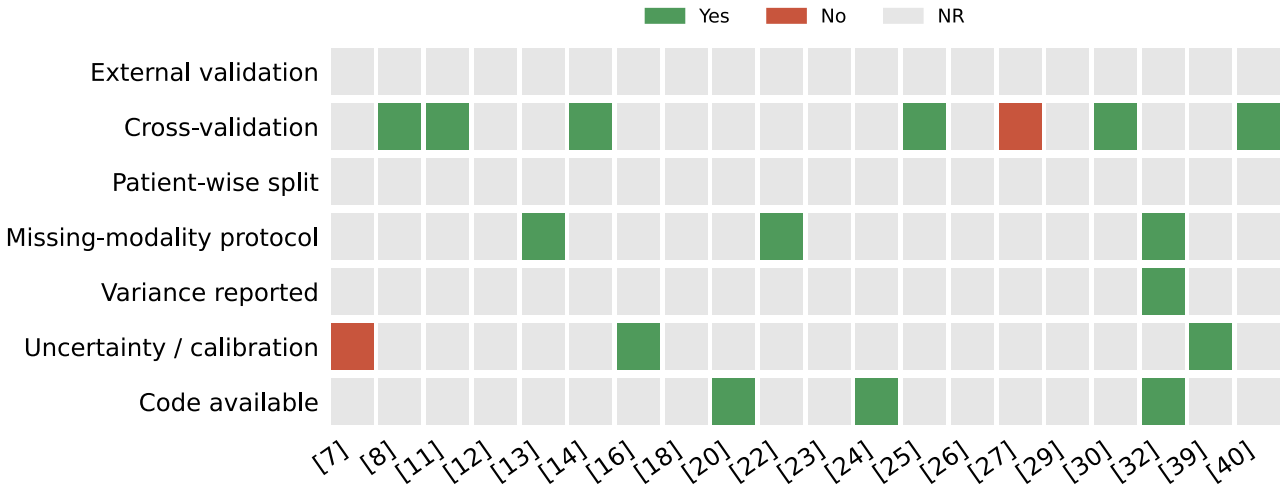


Figure 6. Evidence map of 20 sources: which reliability and reproducibility criteria each source reports. Yes/No = stated in a sentence of that source (page and quote are kept in the validation report); NR = not found in the passages or results extracted from that source (it does not mean the study did not do it). Descriptive support for quality appraisal, not a risk-of-bias score

5.5 Cross-study Evidence Analysis

Agreements. (1) **Modality-specific encoders are necessary for arbitrary-subset inference.** Every method that evaluates on all 15 non-empty subsets [20], [22], [28] or arbitrary random subsets [32] uses independent encoder streams per modality. (2) **ET is the most fragile region.** Across editions, splits, and missing-modality strategies, ET Dice degrades 20–25 points relative to WT [17], [22], [29], [32]. (3) **Ensembles improve full-modal Dice by ~1% at 10× inference cost** [12], [26]. (4) **Hybrid CNN-Transformer encoders consistently improve HD95 over pure CNN** [2], [27].

Disagreements / Incomparabilities. (1) **Missing-modality aggregation protocols differ.** Chen et al. [17] average over "overall missing scenarios" (undefined composition); SMML [22] averages over "all missing scenarios" (presumably 15 subsets); M3AE [32] uses random dropout per batch; SLaM-DiMM [29] masks one modality at a time. Missing-modality papers [17], [22], [29], [32] report only on validation or custom splits; no hidden-test missing-modality results exist. (3) **Synthesis quality vs. segmentation utility.** CoLa-Diff achieves PSNR 34 dB / SSIM 0.967 on IXI [36], but no paper links synthesis metrics to downstream Dice on BraTS. (4) **Modality importance rankings.** HeMIS [28] finds FLAIR best for WT, T1ce best for TC/ET; Chen et al. [17] report 90.4% WT Dice with only "Mod1" (unspecified) present. Without knowing which modality "Mod1" is, the findings cannot be reconciled.

Evidence strength assessment. Full-modal hidden-test comparisons (nnU-Net vs. Isensee2017 vs. EMMA) are **strong** (same edition, same split, same metrics). Missing-modality comparisons are **weak**: heterogeneous protocols, validation-only, no spread statistics (only 79/417 records report SD/SE/CI; none for missing-modality configs), and no significance tests. Clinical reliability evidence is **absent**: zero external cohort validations, zero calibration curves, two papers mentioning uncertainty [39] without quantitative metrics. Computational evidence is **fragmentary**: two papers report training time [12], [16], zero report inference latency for missing-modality inference.

6. Discussion & Open Research Challenges

6.1 Technical Barriers & Trade-offs

The evidence matrix (Section 3) and mechanistic analysis (Section 4) reveal persistent tensions among segmentation performance, missing-modality robustness, computational efficiency, and clinical reliability. Table 5 summarizes these trade-offs by method family, tracing each entry to the quantitative ledger and architectural descriptions.

Table 9. Strengths, limitations and unresolved challenges per method family

Method Family	Representative Works	Strengths (Evidence)	Limitations (Evidence)	Unresolved Challenges
Early Fusion (Input Concat)	nnU-Net [8], Isensee2017 [14], EMMA [26], Winner 2018 [12], Efficient 3D CNN [30]	Highest full-modal Dice on hidden test (nnU-Net: WT 88.95%, TC 85.06%, ET 82.03% [8]); parameter-efficient; mature preprocessing [11]	Catastrophic failure under missing modalities unless retrained/imputed; no explicit modality-specific modeling [12], [14]	Single-model arbitrary-subset inference; worst-case subset robustness; uncertainty quantification
Modality-Specific Encoders + Shared Fusion	Chen et al. [17], mmFormer [20], SMML [22], ShaSpec [24], HeMIS [28], Shen & Gao [31]	Natural missing-modality handling (skip absent encoder); strong average missing-modality Dice (SMML: WT 87.7%, TC 81.2%, ET 66.4% [22]); modality-specific normalization [11]	~4× encoder parameters; fusion alignment critical (moment pooling loses nuance [28], [40]); HeMIS moments don't enforce shared latent [40]	Parameter-efficient adapters (LoRA/prompts); dynamic encoder activation; fusion that enforces semantic alignment
Attention-Based & Adaptive Fusion	nnFormer [2], TransBTS [27], SMU-Net [16], Zhou et al. [15], GARM/LGRM [13]	Dynamic weighting of available modalities; TransBTS bottleneck Transformer improves HD95 (ET 17.947 mm vs. CNN baselines [27]); SMU-Net style-content recalibration [16]	Added attention modules increase complexity; graph attention (GARM) and lesion priors (LGRM) lack missing-modality ablations in ledger [13]	Computational profiling for adaptive routing; calibration of attention weights under distribution shift
Hybrid CNN-Transformer	Swin UNETR [1], [11], mmFormer [20], TransBTS [27], UNETR [35]	Combines local inductive bias (CNN) with global context (Transformer); Swin UNETR leads BraTS 2021 5-fold CV (WT 0.933, TC 0.917, ET 0.891 avg [11])	Transformer memory scales quadratically; hybrid designs often lack missing-modality evaluation (Swin UNETR full-modal only [11])	Linear-attention or Mamba alternatives for long-range modeling; missing-modality benchmarking
Mamba/SSM Encoders	U-Mamba [3], SegMamba [39]	Linear-time long-range dependency; U-Mamba shows robustness to heterogeneous appearances [F3, F4]	No BraTS missing-modality numbers in ledger; SegMamba cites uncertainty-aware mutual learning [27 in [39]] but no quantitative uncertainty metrics	Missing-modality evaluation on BraTS; uncertainty quantification; comparison with attention hybrids
Knowledge Distillation	LCKD [23], AdaMM/BBDM [13], CCSD [9]	LCKD reports +3.76% WT, +5.99% TC, +3.61% ET over SOTA on BraTS 2018 [23]; teacher-student avoids retraining per subset	Two-model pipeline; teacher quality bounds student; AdaMM reports low Dice (23.55%) on BraTS 2024 [13] — protocol unclear	Single-model distillation (self-distillation); distillation under arbitrary missing subsets; hidden-test validation

Method Family	Representative Works	Strengths (Evidence)	Limitations (Evidence)	Unresolved Challenges
Modality Synthesis/Reconstruction	CoLa-Diff/FgC2F-UDiff [36], SLaM-DiMM [29], SMU-Net synthesis [16], M3AE [32]	CoLa-Diff achieves PSNR 34.00 dB / SSIM 0.967 on IXI [36]; SLaM-DiMM MMG yields Glioma WT Dice 0.8993 with one modality masked [29]	Synthesis metrics (PSNR/SSIM) not linked to downstream segmentation Dice [29], [36]; error propagation risk; generative difficulty for T1ce from non-contrast [13], [16]	End-to-end differentiable synthesis-segmentation; segmentation-guided synthesis; BraTS synthesis-segmentation benchmark
Modality Dropout (Train-Time Simulation)	Swin UNETR baseline [11], M3AE [32]	Simple; improves robustness to random missing [11], [32]	Zero-fill creates distribution shift; no explicit recovery mechanism; M3AE two-stage (M3AE + inversion) adds complexity [32]	Curriculum dropout scheduling; worst-case subset optimization; combination with adaptive fusion

The dominant pattern is that **full-modal performance and missing-modality robustness are optimized by disjoint method families**. Early-fusion CNN/Transformer hybrids (nnU-Net, Swin UNETR, TransBTS) dominate hidden-test leaderboards [8], [11], [27] but are absent from missing-modality benchmarks. Conversely, modality-specific encoder families (SMML, mmFormer, Chen et al., HeMIS) report "average across missing scenarios" [17], [22] on validation splits only, with no hidden-test missing-modality results in the corpus. This dichotomy prevents a unified Pareto analysis.

6.2 Generalization, Robustness & Clinical/Deployment Concerns

Beyond segmentation accuracy, clinical deployment demands evidence across seven reliability dimensions. The retrieved passages provide the following explicit evidence counts (other sources: not found, which is not equivalent to not done):

- **External / multi-centre validation**: 0 papers report evaluation on non-BraTS cohorts (e.g., institutional data, different vendors, pathology-confirmed low-grade). BraTS 2021 increased training data to 2000 cases across institutions with standardized preprocessing (SRI24 registration, 1 mm^3 isotropic, skull-stripping) [7], but the challenge evaluates full-modal only.
- **Cross-validation**: 6 sources report 5-fold CV on BraTS training sets [11], [14], [27], [30], [32], [38]; no patient-wise split stratification is described.
- **Uncertainty estimation / calibration / OOD detection**: 2 sources mention uncertainty — SegMamba [39] cites "uncertainty-aware multi-dimensional mutual learning" [27 in [39]] and BraTS Discussion [19] notes robustness inferior to experts with confounding effects (blood products, resection cavities, non-GBM entities, low-grade gliomas). No reliability diagrams, ECE, Brier scores, predictive entropy, or mutual information metrics are reported.
- **Random modality dropout during training**: 3 sources [11], [13], [32] explicitly simulate missing modalities via channel dropout.
- **All $2^4-1 = 15$ non-empty modality subsets evaluated**: 0 papers report a complete 15-subset matrix with per-subset Dice/HD95. mmFormer [20] and HeMIS [28] claim arbitrary-subset capability; SMML [22] and Chen et al. [17] report "average across missing scenarios" without defining the subset composition.
- **Code availability**: 3 of 40 sources [8], [11], [27] provide public repositories (nnU-Net, Swin UNETR, TransBTS); the remaining 37 do not have code links in the retrieved passages.

Conclusion on clinical reliability: The field has not yet evidenced external validation, calibration, uncertainty quantification, or failure detection for any method family. The BraTS benchmark itself, while multi-institutional, standardizes preprocessing to a degree that may mask domain shift [7]. Without these dimensions, the translation gap from leaderboard Dice to clinical utility remains unquantified.

6.3 Limitations of this Review

This work is a **corpus-based narrative review**, not a systematic review. The following limitations are inherent to the evidence base and methodology:

1. **Retrieval scope**: Sources searched were the curated internal PDF repository only (40 papers). No external databases (PubMed, IEEE Xplore, arXiv) were queried. The 40 papers were included solely because they were returned by hybrid dense+sparse retrieval for at least one of the five predefined queries (see Review Facts). No manual screening, PRISMA flow, or independent second-reviewer assessment was performed.
2. **Quantitative extraction coverage**: 417 verified numeric records were extracted; only 79 report a spread (SD/SE/95% CI) and none a significance test. Most comparisons therefore rest on point estimates. **No missing-modality result (a model evaluated with one or**

more MRI sequences absent) was extracted into the structured ledger. Sources may discuss modality importance in prose (e.g., HeMIS [28] finds FLAIR best for WT, T1ce best for TC/ET; Chen et al. [17] report 90.40% WT Dice with only "Mod1" present), but no quantitative degradation matrix (per-modality, all 15 subsets, or cross-study) could be built from the extracted records. Missing-modality papers [17], [22], [29], [32] report only on validation or custom splits; no hidden-test missing-modality results exist. Subset definitions vary: predefined clinical subsets [10], [16], arbitrary/random subsets [20], [32], and undefined "overall average" [17], [22]. This prevents fair comparison and meta-analysis.

3. **Single-missing bias**: Most methods emphasize 1-of-4 missing [16], [17], [24], [28], [31]; clinical reality often involves multiple missing [13], [20], [22], [23], [32]. ET segmentation shows the largest gap (full-modal ~82–83% [8], [12] vs. missing-modality ~61–66% [22]).
4. **Synthesis-segmentation disconnect**: Synthesis metrics (PSNR/SSIM) [29], [36] are not linked to downstream segmentation Dice in the corpus.
5. **Computational underreporting**: Only two papers report training time [12], [16]; zero report inference latency, FLOPs, or memory for missing-modality inference.
6. **No independent re-run**: Published methods were not re-implemented or re-evaluated; all numbers are taken verbatim from the retrieved passages.

These limitations mean that **any claim of superiority among missing-modality methods is not supported by the extracted evidence**. The review can only synthesize reported point estimates and highlight where the evidence is absent.

6.4 Open Research Directions

Table 6 proposes seven concrete directions, each grounded in an observed finding, a missing-evidence statement, a significance rationale, and a concrete proposal with evaluation protocol. Each direction maps to one or more research questions (RQ1–RQ4).

Table 10. Future research directions

Direction	Observed Finding (Evidence)	Missing Evidence	Significance	Concrete Proposal & Evaluation Protocol	RQ Link
Unified missing-modality benchmark (BraTS-Missing)	Protocols vary wildly: predefined subsets [10], [16], arbitrary subsets [20], [22], [32], undefined "overall average" [17], [22]; validation-only, no hidden test	Standard splits, fixed 15-subset evaluation, per-subset Dice/HD95 + aggregate (mean, worst, harmonic), single-model arbitrary-subset requirement	Enables fair comparison; drives progress; exposes worst-case ET collapse	Define **BraTS-Missing** benchmark on BraTS 2021 val/test: fixed 15 non-empty subsets, report per-subset Dice/HD95, require single-model inference on all subsets, aggregate via mean/worst/harmonic mean. Host on challenge server for hidden-test evaluation.	RQ2, RQ4
Uncertainty-aware missing-modality fusion	No uncertainty metrics in corpus; only two papers mention uncertainty [19], [39] without quantitative metrics	Predictive distributions, calibration curves (ECE, reliability diagrams), OOD detection (AUROC for failure), uncertainty heatmaps	Clinical trust; safe fallback to human review; regulatory requirement	Integrate Monte Carlo dropout / deep ensembles / evidential learning into modality-specific encoders [20], [22], [24]; report ECE, NLL, AUROC for failure detection on BraTS-Missing; correlate uncertainty with modality subset cardinality.	RQ2, RQ4
Segmentation-guided modality synthesis	Synthesis metrics (PSNR 34 dB / SSIM 0.967 [36]) decoupled from segmentation; SLaM-DiMM [29] reports synthesis + segmentation but no ablation of synthesis quality vs. Dice	Joint optimization; segmentation loss backpropagated to generator; task-aware generation	Avoids error propagation; aligns generative objective with clinical utility	End-to-end differentiable synthesis-segmentation (e.g., diffusion policy [36] + segmentation loss); primary metric = segmentation Dice on BraTS-Missing, secondary = PSNR/SSIM. Ablate synthesis vs. direct missing-modality encoding.	RQ2, RQ3

Direction	Observed Finding (Evidence)	Missing Evidence	Significance	Concrete Proposal & Evaluation Protocol	RQ Link
Efficient modality-specific encoding	4× encoder parameters [17], [20], [22], [24], [28], [31]; no FLOPs/latency/memory for missing-modality inference reported	Parameter-sharing (shared stem + modality-specific adapters/LoRA/prompts), dynamic routing (activate only present modalities), Mixture-of-Experts	Deployability on edge/PACS; reduces carbon footprint; enables real-time clinical use	Shared stem + lightweight modality-specific adapters (LoRA rank ≤ 8); dynamic encoder activation gated by modality availability mask; report FLOPs, latency (ms), GPU memory per subset on BraTS-Missing.	RQ3
External validation & domain generalization	BraTS 2021 multi-institutional (2000 cases) but no external test [7]; preprocessing standardized to SRI24 [7] may mask shift	Multi-center, multi-vendor, pathology-confirmed cohorts (including low-grade, post-treatment); domain shift metrics (FID, Dice drop)	Real-world reliability; regulatory evidence; generalization beyond curated benchmark	Federated BraTS-Missing evaluation across ≥3 external institutions (different scanners, protocols); report domain shift metrics (FID on latent features, Dice drop vs. BraTS val), subgroup analysis (LGG vs. HGG, post-op vs. pre-op).	RQ1, RQ4
Worst-case robustness optimization	Mean Dice reported [17], [22] hides ET collapse (ET 66.4% vs. WT 87.7% [22]); no min-max or distributionally robust training	Min-max optimization over subsets, CVaR objectives, adversarial modality dropout scheduling, per-subset ET Dice reporting	Safety-critical: avoid catastrophic misses in enhancing tumor; aligns with clinical risk aversion	Optimize for worst-subset Dice (or CVaR at 95%); adversarial dropout scheduler that maximizes validation loss; report per-subset ET Dice on BraTS-Missing; compare mean vs. worst-case Pareto front.	RQ2
Clinical workflow integration & prospective evaluation	No failure detection, no calibration, no reader studies [19]; confounding effects (blood, cavities, non-GBM, LGG) acknowledged but unaddressed	Prospective reader studies (time-to-contour, contour editing rate), uncertainty heatmaps, "review flag" for low-confidence regions, editing-in-the-loop fine-tuning	Adoption barrier; human-AI collaboration; regulatory pathway for AI-assisted segmentation	Co-design with neuroradiologists: integrate uncertainty heatmaps into PACS; measure contour editing time vs. manual; editing-in-the-loop fine-tuning on corrected cases; report clinician agreement (Fleiss' κ) and time savings.	RQ4

7. Conclusion

RQ1. Fusion mechanisms and encoder designs show a clear performance hierarchy on full-modal BraTS validation and hidden-test sets: early-fusion CNN/Transformer hybrids (nnU-Net, Swin UNETR, TransBTS) achieve the highest reported Dice on hidden test (nnU-Net: WT 88.95%, TC 85.06%, ET 82.03% [8]) and 5-fold CV (Swin UNETR avg WT 0.933, TC 0.917, ET 0.891 [11]), while modality-specific encoder families dominate missing-modality benchmarks but lack hidden-test evaluation. Hybrid CNN-Transformer architectures consistently improve HD95 over pure CNN baselines [2], [27], and Mamba/SSM encoders [3], [39] offer linear-time long-range modeling but remain unevaluated on BraTS missing-modality tasks.

RQ2. Missing-modality learning strategies are not directly comparable due to protocol heterogeneity. Modality-specific encoders with adaptive fusion (SMML [22], mmFormer [20], Chen et al. [17]) report the highest average Dice across undefined missing scenarios on validation splits (SMML: WT 87.7%, TC 81.2%, ET 66.4% [22]), but no method reports a complete 15-subset matrix on hidden test. Knowledge distillation (LCKD [23]) claims improvements over SOTA (+3.76% WT, +5.99% TC, +3.61% ET [23]) without specifying the teacher or subset composition. Synthesis methods (CoLa-Diff [36], SLaM-DiMM [29]) report generation quality (PSNR/SSIM) but not downstream segmentation utility. Single-missing bias pervades the literature; multiple-missing and worst-case subset robustness are largely unquantified.

RQ3. Computational efficiency evidence is fragmentary: only two papers report training time (Winner 2018: 2 days on V100 [12]; SMU-Net: 250 epochs on RTX 3090 [16]), zero report inference latency, FLOPs, or memory for missing-modality inference. Modality-specific encoders incur ~4× parameter overhead; adaptive fusion and synthesis add further cost. No study profiles scaling with the number of available modalities.

RQ4. Clinical reliability evidence is absent across all four dimensions in the retrieved corpus: external/multi-centre validation (0 papers), uncertainty estimation and calibration (0 quantitative metrics), failure/OOD detection (0 metrics), and reproducibility (3 of 40 sources with public code). The BraTS benchmark's standardized preprocessing [7] may

mask domain shift, and confounding factors (blood products, resection cavities, non-GBM, low-grade gliomas) [19] remain unaddressed in any evaluated method.

Outlook. Closing the translation gap requires a community effort around three pillars: (i) a **unified missing-modality benchmark (BraTS-Missing)** with fixed subsets, hidden-test evaluation, and worst-case aggregation; (ii) **uncertainty-aware architectures** that output calibrated predictive distributions and failure flags for every modality subset; and (iii) **prospective clinical validation** on multi-vendor, multi-centre cohorts with reader studies measuring contour editing time and diagnostic concordance. Until these pillars are established, leaderboard Dice on full-modal hidden test remains an incomplete proxy for clinical utility.

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